

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Artificial Intelligence

Semester: Spring

Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Differentiate between Intelligence and Artificial Intelligence. What are the application areas of AI? State in brief. 8
- b) You are given two jugs, a 4-gallon one and a 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug? Solve by production rule system. 7
- a) What is state space search? Given any example of a game which happens to be a problem of state space search and justify your answer properly. 8
- b) Explain Min-max search. Explain the advantages of Alpha-Beta Pruning over Min-max search? 7
- a) Compare depth limited search and iterative deepening search in terms of time and space complexity, completeness and optimality. 8
- b) Differentiate propositional and a predicate statement. How does first order predicate logic extend propositional logic to overcome its limitations? 7
- a) What are the problems with hill climbing and how can they be solved? Define plateau 7
- b) Represent the following sentences in first order logic: 8
 - i. Nabin is intelligent than all other students in his class.
 - ii. Neither Sabin nor Kabin is unhappy.
 - iii. Some cats are domestic pets
 - iv. Some turtles are faster than rabbits.
- a) Explain how statistical reasoning aids in inference and reasoning in light of Bayes theorem. 7

b) Differentiate semantic nets and frame based knowledge representation. 8

a) State and explain in brief about EBL-system. What are its advantages 7

b) Explain the different steps involved in the natural language processing (NLP) with suitable block diagram and examples. 8

Write short notes on: (*Any two*) 2×5

a) Turing test

b) Conceptual dependencies

c) Expert System

